Scatter Plots and Bivariate Association — Direction, Form, and Strength

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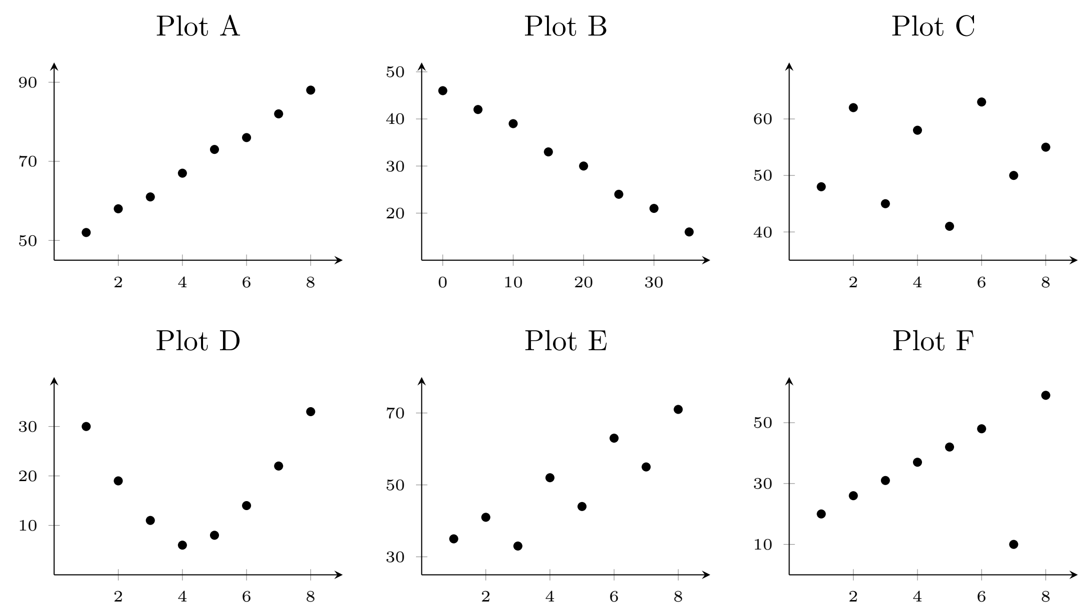
Describing a scatter plot in three words: which way it goes, what shape it makes, and how tightly the points hug the trend

Directions:

- Six scatter plots, **Plot A** to **Plot F**, are printed below. Every problem names the plot it is about.
- A complete description of a scatter plot has three parts: **direction** (positive, negative, or none), **form** (linear or nonlinear), and **strength** (how close the points sit to the trend).
- Each problem asks you to compute something first and describe second. The number is the evidence; the description is the answer.
- "Residual" means *actual y minus predicted y*. A residual close to zero means that point sits close to the line.

The six plots



Trend lines you will need: Plot A follows $y = 5x + 47$. Plot E follows $y = 5x + 25$.

1. Plot A — which way does it go?

(a) Write the y-value of the leftmost point and the y-value of the rightmost point, then put $>$, $<$, or $=$ between them.

leftmost _________ ______ rightmost _________

(b) State the direction of the association in Plot A, and say how your comparison in (a) supports it.

2. Plot B — the other direction.

(a) Do the same thing: leftmost y-value, rightmost y-value, and the correct symbol between them.

leftmost _________ ______ rightmost _________

(b) State the direction, and complete this sentence: "as x increases, y tends to ____________".

3. Plot D — what shape do the points make?

(a) Which x-value carries the *lowest* y-value in Plot D?

(b) Follow the points from left to right. Describe in words what they do, then say whether the form is **linear** or **nonlinear**.

Answer: ________________

4. Plots A and E — measuring how far a point sits from the trend. The residual of a point is actual y − predicted y.

(a) In Plot A the point at $x = 5$ has $y = 73$, and the trend line $y = 5x + 47$ predicts $5(5) + 47$. Find the residual.

(b) In Plot E the point at $x = 5$ has $y = 44$, and the trend line $y = 5x + 25$ predicts $5(5) + 25$. Find the residual.

Plot A residual _________ Plot E residual _________

(c) Which of the two points sits closer to its trend line, and how do you know from the two numbers?

5. Plot C — when there is no direction at all.

(a) Which x-value carries the highest y-value? (b) Which carries the lowest?

highest at $x =$ ________ lowest at $x =$ ________

(c) On Plot A the highest and lowest y-values sat at opposite ends of the plot. Where do they sit here? Use that to explain why Plot C has **no association**.

6. Plot F — one point that does not belong.

(a) Which x-value carries the lowest y-value in Plot F? (Careful: the question asks where the lowest *height* is, not the smallest x.)

(b) Cover that point with your finger. Now describe the direction, form and strength of what is left.

(c) A point like this is called an **outlier**. Explain in one sentence what it does to your judgement of the strength of Plot F.

Answer: ________________

7. Plots A and E — putting a number on strength. Here are the residuals of all eight points on each plot, measured from the trend lines printed with the figures.

Plot A residuals	0	1	−1	0	1	−1	0	1
Plot E residuals	5	6	−7	7	−6	8	−5	6

(a) Find the range of each list of residuals (largest minus smallest).

Plot A range _________ Plot E range _________

(b) Both plots have a positive linear association. Use your two ranges to say which one is **stronger**, and explain what the range of the residuals is actually measuring.

8. Plot D again — why "direction" is the wrong question here.

(a) What is the highest y-value anywhere on Plot D?

(b) A student argues: "the last point is higher than the first point, so Plot D shows a positive association." The student's evidence is true. Explain why the conclusion is still wrong, using the word **form**.

Answer: ________________

9. Plot F — what one outlier does to a summary.

(a) Find the mean of all eight y-values in Plot F: $20, 26, 31, 37, 42, 48, 10, 59$.

(b) Now remove the outlier and find the mean of the remaining seven y-values. Round to the nearest hundredth.

with outlier _________ without _________

(c) By how much did one point out of eight move the mean? Say what that shows about trusting a summary from a small data set.

10. **Challenge.** Write two complete descriptions, one for **Plot A** and one for **Plot E**. Each description must use all three words — direction, form, strength — and must give the evidence for each.

Then judge this claim: *"Plot E shows a strong link between the two quantities."* Say whether you agree, and support your answer with the residual numbers from problem 7 rather than with the slope of the line.

Answer: ________________